

# Cooperative Multi-Agent Learning in Overcooked-AI: A Markov Game Perspective on Coordination and Reward Design

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**Abstract:** We propose a project on cooperative multi-agent learning in the Overcooked-AI environment. Our goal is to investigate how reward design affects coordination quality, learning efficiency, and overall task performance in a shared sequential decision-making setting. We formulate the problem as a cooperative Markov game in which multiple agents interact in a common kitchen environment and must coordinate to complete cooking and delivery tasks efficiently. We will compare a baseline setting with shared team rewards against shaped reward variants that provide intermediate incentives for subtasks. All experiments will be conducted in simulation, and we will evaluate the learned policies using task return, completed dishes, and qualitative coordination behavior.

**Keywords:** multi-agent reinforcement learning, Overcooked-AI, Markov games

## 1 Introduction

In this project, we study cooperative multi-agent decision making in the Overcooked-AI environment. Overcooked-AI provides a structured simulation setting in which two agents must collaborate under shared constraints to prepare and deliver dishes. This project is motivated by both robot learning and game-theoretic perspectives. From the robot learning perspective, the environment serves as a sequential decision-making benchmark for learning cooperative policies. From the game-theoretic perspective, the task can be modeled as a cooperative Markov game in which the payoff structure influences the behavior that emerges during repeated interaction.

Our main research question is whether different reward designs lead to different levels of coordination efficiency, learning stability, and final task performance. In particular, we will compare a sparse team-reward setting with reward shaping variants that provide intermediate rewards for subtasks such as picking up ingredients, placing ingredients into pots, and serving dishes. Through these comparisons, we aim to understand whether additional incentives can improve both learning speed and the quality of emergent coordination.

## 2 Simulator, Dataset, Repository, and Platform

We will use **Overcooked-AI** as the main simulation platform for this project. Overcooked-AI is a widely used benchmark for cooperative multi-agent learning, where two agents must coordinate in a grid-based kitchen environment to accomplish shared cooking tasks. The platform includes multiple layouts with varying coordination difficulty and supports controlled evaluation of policy performance.

Our implementation will be based on **Python** and **PyTorch**, using the official Overcooked-AI code-base for environment simulation and evaluation. We plan to begin with a parameter-sharing rein-

forcement learning baseline and then compare alternative reward formulations. Since the project is entirely simulation-based, we will not use any physical robotic hardware or real-world dataset.

### 3 Division of Work

We divide the project into two main components: environment setup and experimentation, and learning algorithm implementation and analysis.

Junzhe Huang will focus primarily on environment setup, including installing and validating Overcooked-AI, selecting representative layouts, running baseline agents, and preparing rollout visualization tools for qualitative analysis.

Xingzhi Qian will focus primarily on implementing and training learning-based agents, including building the reinforcement learning pipeline, tuning hyperparameters, and logging training performance across different reward settings.

Both team members will jointly work on the Markov game formulation, reward design, experiment design, debugging, result interpretation, and the writing of the final report and presentation video.

### 4 Timeline and Progress Report Targets

Our project timeline is organized into four stages.

**Week 1:** We will finalize the project scope, install and test the Overcooked-AI environment, select one or two layouts, and run simple baseline agents to verify that the environment and evaluation pipeline are working correctly.

**Week 2:** We will implement the baseline learning method and conduct initial training experiments. At this stage, we expect to obtain preliminary training curves and verify that the agents can learn at least partial cooperative behavior.

**Progress Report Target:** By the time of the progress report, we aim to have:

- a working Overcooked-AI simulation setup,
- baseline agent results,
- an initial trainable reinforcement learning pipeline, and
- preliminary comparisons between random or scripted behavior and a learned policy.

**Week 3:** We will implement and test multiple reward designs, including shared sparse reward and shaped reward variants, and compare their effects on convergence speed, final return, and coordination quality.

**Week 4:** We will run final experiments, organize both quantitative and qualitative results, and prepare the final report and presentation video. We will also refine the framing so that the final project clearly connects cooperative reinforcement learning with Markov game analysis.

### 5 Difficulties

Several challenges may arise during the course of this project. First, cooperative multi-agent learning is often less stable than single-agent learning because each agent must adapt not only to the environment but also to the evolving behavior of its teammate. This non-stationarity may make training less predictable and more difficult to tune.

Second, reward shaping may improve sample efficiency, but it may also unintentionally bias the learned policy toward short-term subgoals rather than effective long-term cooperation. As a result, designing intermediate rewards that encourage coordination without distorting the overall task objective will be an important challenge.

Third, policy evaluation in cooperative environments is not straightforward. Even when quantitative metrics such as return or completed dishes improve, it may still be difficult to determine whether the agents have developed robust coordination strategies or are merely exploiting layout-specific patterns. For this reason, careful qualitative analysis of agent behavior will also be necessary.

## 6 Conclusion

This project studies cooperative multi-agent learning in a controlled simulation environment and connects reinforcement learning with a Markov game perspective on coordination. By comparing different reward structures in Overcooked-AI, we aim to better understand how incentive design shapes emergent cooperative behavior, learning stability, and final task performance. We expect this project to provide a clear simulation-based case study of how reward design influences coordination in multi-agent systems.

**Group ID:** To be added after receiving proposal feedback. We will include this ID number in the final report and project presentation video as required.